Συστήματα Διαχείρισης και Ανάλυσης Δεδομένων
Διδάσκων: Ιωάννης Κωτίδης

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Λύσεις Δεύτερης Εργασίας

Ζήτημα 1 - Λύση

```
/* Δημιουργία του πίνακα accdata και μαζική εισαγωγή δεδομένων */

create table accdata
(accident_id varchar(15),
 severity_id int,
 severity varchar(10),
 road_surface_conditions_id int,
 road_surface_conditions varchar(50),
 accident_date date,
 number_of_vehicles int,
 vehicle_type_id int,
 vehicle_type varchar(50),
 driver_class_id int,
 sex_of_driver varchar(6),
 age_of_driver int,
 sex_of_casualty varchar(6),
 age_of_casualty int
 );

BULK INSERT accdata
FROM 'C:\DATAWAREHOUSE\ACCDATA.txt'
WITH (FIRSTROW =2, FIELDTERMINATOR='|', ROWTERMINATOR = '\n');

/* Δημιουργία Σχήματος Αποθήκης Δεδομένων */

Create table severity
 (severity_id int primary key,
  severity varchar(10)
)
insert  severity select distinct severity_id, severity from accdata

create table road_surface
(road_surface_conditions_id int primary key,
 road_surface_conditions varchar(50),
 );
insert  road_surface select distinct road_surface_conditions_id,
road_surface_conditions from accdata

create table vehicle
(vehicle_type_id int primary key ,
 vehicle_type varchar(50),
 );
insert  vehicle select distinct vehicle_type_id, vehicle_type from accdata
```

```
create table driver_class
  (driver_class_id int primary key,
   sex_of_driver varchar(6),
   age_of_driver int
   );

insert driver_class
   select distinct driver_class_id, sex_of_driver, age_of_driver from accdata


create table d_time
 (
   accident_date date primary key,
   t_year int,
   t_quarter int,
   t_month int

)

insert into d_time
    select distinct accident_date, datepart(year, accident_date),
             datepart(quarter, accident_date),datepart(month, accident_date
        from accdata

create table accidents
(accident_id varchar(15) primary key,
 severity_id int foreign key references severity(severity_id),
 road_surface_conditions_id int foreign key references
road_surface(road_surface_conditions_id),
 vehicle_type_id int foreign key references vehicle(vehicle_type_id),
 driver_class_id int foreign key references driver_class(driver_class_id),
 accident_date date foreign key references d_time(accident_date),
 number_of_vehicles int,
 number_of_casualties int
 );

insert accidents
    select accident_id, severity_id, road_surface_conditions_id,
vehicle_type_id, driver_class_id, accident_date,
            number_of_vehicles, count(*)
              from accdata
    group by accident_id, severity_id, road_surface_conditions_id,
vehicle_type_id, driver_class_id, accident_date,
            number_of_vehicles
```

Ζήτημα 2 - Λύση

```
--Q1

select t_year, severity, count(*)
   from severity, accidents, d_time
where severity.severity_id=accidents.severity_id and
      d_time.accident_date=accidents.accident_date
   group by t_year, severity
   order by t_year desc


--Q2
select sex_of_driver, age_of_driver,
       count(accidents.accident_id) as NUMBER_OF_ACCIDENTS,
       sum(number_of_casualties) as NUMBER_OF_CASUALTIES
   from driver_class, accidents, severity
where severity.severity_id=accidents.severity_id and
      driver_class.driver_class_id=accidents.driver_class_id and
         severity='Fatal'
   group by sex_of_driver, age_of_driver
   order by sex_of_driver, age_of_driver



 --Q3
 select road_surface_conditions, severity, count(accidents.accident_id)
  from road_surface, severity, accidents
  where road_surface.road_surface_conditions_id=accidents.road_surface_conditions_id and
        severity.severity_id=accidents.severity_id
    group by  road_surface_conditions, severity
    order by road_surface_conditions, severity

--Q4
select t_year, vehicle_type,
       count(accidents.accident_id) AS NUMBER_OF_ACCIDENTS,
       sum(number_of_casualties) AS NUMBER_OF_CASUALTIES
   from vehicle, accidents, d_time
   where vehicle.vehicle_type_id=accidents.vehicle_type_id and
         d_time.accident_date=accidents.accident_date and
          number_of_vehicles > 2
     group by t_year, vehicle_type
       order by t_year, vehicle_type

--Q5
select t_year, t_quarter, t_month,
       count(accidents.accident_id) AS NUMBER_OF_ACCIDENTS,
       sum(number_of_vehicles) AS NUMBER_OF_VEHICLES,
       sum (number_of_casualties) AS NUMBER_OF_CASUALTIES
    from d_time, accidents
where d_time.accident_date=accidents.accident_date
group by rollup(t_year, t_quarter, t_month)
```

Ζήτημα 3 - Λύση

1.

```
select severity, road_surface_conditions, vehicle_type,
       count(accidents.accident_id) as NUMBER_OF_ACCIDENTS
  from severity, road_surface, vehicle, accidents
 where severity.severity_id = accidents.severity_id and
       road_surface.road_surface_conditions_id=accidents.road_surface_conditions_id and
      vehicle.vehicle_type_id=accidents.vehicle_type_id
  group by cube (severity, road_surface_conditions, vehicle_type)
```

2.

```
create view V1
with schemabinding as
select [s].severity, [r].road_surface_conditions, [v].vehicle_type,
       count_big (*) as NUMBER_OF_ACCIDENTS
   from dbo.severity [s], dbo.road_surface [r],
           dbo.vehicle [v], dbo.accidents [a]
   where [s].severity_id = [a].severity_id and
         [r].road_surface_conditions_id=[a].road_surface_conditions_id and
         [v].vehicle_type_id=[a].vehicle_type_id
  group by [s].severity, [r].road_surface_conditions, [v].vehicle_type


  create unique clustered index idx1 on
                       v1(severity, road_surface_conditions, vehicle_type)

  /* group by severity, road_surface_conditions */
  select severity, road_surface_conditions, sum(NUMBER_OF_ACCIDENTS)
       from v1
    group by severity, road_surface_conditions


 /* group by severity, vehicle_type */
  select severity, vehicle_type, sum(NUMBER_OF_ACCIDENTS)
       from v1
    group by severity, vehicle_type

  /* group by road_surface_conditions, vehicle_type */
    select road_surface_conditions, vehicle_type, sum(NUMBER_OF_ACCIDENTS)
       from v1
    group by road_surface_conditions, vehicle_type

 /* group by severity */
  select severity, sum(NUMBER_OF_ACCIDENTS)
       from v1
    group by severity

/* group by  vehicle_type */
  select  vehicle_type, sum(NUMBER_OF_ACCIDENTS)
       from v1
    group by  vehicle_type

/* group by road_surface_conditions */
  select road_surface_conditions, sum(NUMBER_OF_ACCIDENTS)
       from v1
    group by road_surface_conditions

/* group by none */
select sum (NUMBER_OF_ACCIDENTS) from V1
```